SVKM'S NMIMS

MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT & ENGINEERING

Academic Year: 2025-2026

Program: B.Tech Stream: Mechatronics	Year: 2nd Semester: IV
Subject: Dynamic Systems Modeling (702MH0C023)	Time: 45 Minutes
Date:	No. of Pages: 1
Marks: 10	Set: 2

Re-Examination

Instructions: Candidates should read carefully the instructions.

1) Figures in brackets on the right hand side indicate full marks.
2) Syllabus scope: Unit 1 – Unit 5 (complete).
3) Assume suitable data if necessary.
4) Question 1 is compulsory.
5) Answer any 2 from the remaining questions.

Q1 Answer briefly:

CO- 1; SO- 1; BL- 1,2

a. (i) Write the expressions for the radial and transverse components of acceleration in polar coordinates (a_r and a_θ) and identify the centripetal term. [2]
(ii) State Hamilton's principle (principle of least action) and explain its relationship to the Euler-Lagrange equation.
(iii) Differentiate between holonomic and non-holonomic constraints. Give one example of each from a mechanical system.
(iv) State the parallel axis theorem for moment of inertia and give its mathematical expression.

CO- 1,2; SO- 1,2; BL- 2,3

b. (i) A particle moves along a spiral with r = 3 m, $\dot{r} = 0$, and $\dot{\theta} = 2$ rad/s (constant). Calculate the centripetal acceleration and state its direction. [2]
(ii) Explain the concept of generalized coordinates. For a double pendulum in a plane, state the number of DOF and identify the generalized coordinates.
(iii) Define the Coriolis acceleration. State the expression for its magnitude and explain in which physical situations it becomes significant.
(iv) A uniform rod of mass m = 3 kg and length L = 1.2 m is pivoted at one end. Calculate its moment of inertia about the pivot using the parallel axis theorem.

Q2
CO- 1,2; SO- 1,2; BL- 3

A simple pendulum consists of a point mass m = 2 kg attached to a massless, inextensible string of length L = 0.5 m. The bob swings in a vertical plane under gravity (g = 9.81 m/s^2). Using the angle θ from the vertical as the generalized coordinate, analyze the system using the Lagrangian method. [3]
Use the following relations: Lagrangian L = T − V; Euler-Lagrange equation: $d/dt(\partial L/\partial \dot{q}) - \partial L/\partial q = 0$; for small angles: $\sin\theta \approx \theta$.
(a) State the number of degrees of freedom of this system and identify the generalized coordinate.
(b) Write the position of the mass in Cartesian coordinates (x, y) in terms of θ and L.
(c) Derive the kinetic energy T in terms of θ, $\dot{\theta}$, m, and L.
(d) Write the potential energy V in terms of θ, m, g, and L.
(e) Form the Lagrangian L = T − V and apply the Euler-Lagrange equation to derive the equation of motion.
(f) For small oscillations ($\sin\theta \approx \theta$), determine the natural frequency of the pendulum.

Q3
CO- 2; SO- 2; BL- 3,4

A horizontal turntable rotates at a constant angular velocity Ω = 5 rad/s about a vertical axis. A small ball at r = 2 m from the center is pushed radially outward at a constant relative speed v_{rel} = 3 m/s ($\dot{r}$ = 3 m/s, $\ddot{r}$ = 0). The angular acceleration of the turntable is zero (α = 0). [3]
(a) Write the general expression relating the acceleration observed from the fixed (inertial) frame to quantities measured in the rotating frame. Identify each term.
(b) Calculate the centripetal acceleration magnitude at r = 2 m.
(c) Calculate the Coriolis acceleration magnitude.
(d) State the direction of the Coriolis acceleration relative to the radial velocity.
(e) Calculate the total acceleration magnitude of the object as observed from the fixed (inertial) frame.
(f) If the ball continues at the same radial speed, find the centripetal acceleration when r = 4 m and comment on how it changes.

Q4
CO- 2,3;
SO- 2,3;
BL- 4,5

A uniform rod of mass $m = 2$ kg and length $L = 1.2$ m is pivoted at one end and oscillates as a physical (compound) pendulum under gravity. Take $g = 9.81$ m/s^2. [3]

Use: $I_{rod\_cm} = (1/12)mL^2$; parallel axis theorem: $I = I_{cm} + md^2$; period of physical pendulum: $T = 2\pi\sqrt{(I_{pivot} / (mgd))}$, where d = distance from pivot to center of mass.

(a) Calculate the moment of inertia of the rod about its center of mass (I_{cm}).

(b) Using the parallel axis theorem, calculate the moment of inertia about the pivot at one end (I_{pivot}).

(c) Find the distance d of the center of mass from the pivot.

(d) Write the equation of motion for small-angle oscillations and determine the natural frequency ω_n.

(e) Calculate the time period of oscillation T.

(f) Compare this period with that of a simple pendulum of the same length ($L = 1.2$ m) and explain the difference.